

P-TUC 2.0 — Critical Highlights

Purpose

- Ensures **structurally honest neuron outputs** by making **silence the default** for uncertain or contradictory inputs.
- Extends classical architectures by integrating **temporal coherence, predictive error gating, and hierarchical validation**.

Core Mechanisms

1. **Temporal Anchoring (Phase 1)**
 - Neurons output only if inputs remain stable over **integration window** T (~500 ms).
 - Ground State: $y(t) = 0$ if inputs are unstable.
2. **Predictive Error-Gated Phase-Locked Loop (Phase 2)**
 - Prediction error: $\epsilon(t) = |S_{ff}(t) - P_{fb}(t)|$
 - PLL slows neuron clock when ϵ is high $\rightarrow$ prevents unreliable output.
 - Output realized only if $\int_{t-T}^t \epsilon(\tau)^2 d\tau < \tau_{\text{threshold}}$.
3. **Socket-Based Modular Architecture (Phases 3–4)**
 - **Comparator Socket:** Computes ϵ , pluggable with Bayesian, KL, or contrastive metrics.
 - **Gating Socket:** Predictive brake, pluggable with stochastic or quantum-inspired gates.
 - **Plasticity Socket:** Weight updates ($\Delta W = \eta \cdot R \cdot \epsilon$), supports STDP or metaplasticity.
 - **Extensibility:** Modular design allows new gates, learning rules, or error metrics **without affecting stability**.
4. **Hierarchical & Asynchronous Validation (Phase 5)**
 - Layer N clocks constrained by Layer N-1.
 - Fast path $\rightarrow$ familiar inputs; slow path $\rightarrow$ novel inputs.
 - Emergent sparsity: only resonant neurons consume energy.

Failure Modes & Ground State Enforcement (Phase 6–7)

Input Type	P-TUC 2.0 Response	Physical State
Clear Truth	High-Speed Tick	Full Resonance
Ambiguity	Delayed Tick	Evidence Accumulation
Contradiction	Permanent Silence	Clock Freeze

Input Type	P-TUC 2.0 Response	Physical State
Pure Noise	Ground State	Sub-critical Coherence

- Structural mechanisms **prevent hallucinations** and unreliable outputs.
 - Formal analysis shows **convergence to Ground State** for high error, preventing spurious activity.
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Key Advantages

- **Structural Honesty:** Dishonest outputs are **inherently inhibited**, not externally filtered.
 - **Energy Efficiency:** Silent neurons consume minimal energy.
 - **Neuromorphic Compatibility:** Designed for modular, scalable hardware.
 - **Extensible:** Socket interfaces allow seamless integration of new modules or rules.
 - **Predictable Behavior:** Temporal and predictive gating ensures outputs are coherent, traceable, and reliable.
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Comparative Position

- Bridges gap between **Forward-Forward**, **Equilibrium Prop**, and **Active Inference**:
 - Maintains **temporal coherence** (unlike Forward-Forward).
 - Avoids **slow two-phase convergence** (unlike Equilibrium Prop).
 - Implements **energy-efficient, structurally honest gating** (unlike Active Inference).
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Takeaway

P-TUC 2.0 is a **modular, extensible, and theoretically grounded architecture** that guarantees **coherence-driven, honest outputs**, providing a framework suitable for both **neuromorphic hardware** and **future AI research**.